

Ava Kim Cohen

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WORK EXPERIENCE

Acorn Investment Partners (a portfolio company of Oaktree Capital Management, L.P.) **May – August 2026**
Artificial Intelligence Intern *Greenwich, CT*

- Re-architected internal AI applications serving multiple users: cut per-run cost by 59% and eliminated prompt-injection exposure for a daily synthesis bot; built a RAG-based market/news digest with an incremental memory layer, parallelizing ingestion and reducing re-indexing time by 85%.
- Built and deployed an automated Python P&L and performance reporting platform with over 20k lines of code and 871 tests, reconciled cell-exact against the existing manual process and running unattended in production.

Jump Trading **June – July 2025**
Quantitative Research Intern *Chicago, IL*

- Benchmarked LLMs across tasks involving logic, spatial reasoning, and strategy-based games. Results were published in two manuscripts (<https://arxiv.org/abs/2510.20205> and <https://arxiv.org/abs/2510.20198>).
- Developed a recursive agent using Monte Carlo Tree Search and LLM-based runtime code generation, improving 2048 game score by 5x over random baseline.
- Fine-tuned Llama and Qwen models using synthetic datasets to improve LLM responses and behavior.
- Participated in click trading exercises and lectures from employees on asset classes, trading strategies, etc.

Amyoli Internet Research Lab at The University of Chicago **June – August 2023 & 2024**
Student Research Assistant *Chicago, IL*

- Co-authored a Human-Computer Interaction paper on the quantitative effects of dark patterns in subscription video-on-demand platforms, accepted for publication in CSCW 2025 (<https://doi.org/10.1145/3710928>).
- Conducted statistical analysis and generated visualizations using Python (pandas, NumPy, altair).
- Developed a Python script to validate participant data and flag entries that violated study guidelines.

PERSONAL PROJECTS

Idea Generation for Investments **Summer 2025**

- Built an automated investment insight tool using Palantir's Artificial Intelligence Platform (AIP) and LLMs that analyzes news content and extracts investment signals.
- Implemented a web scraper in Python that extracts metadata from recent articles across news outlets.

EDUCATION

Columbia University **New York, NY**
Bachelor of Science, Applied Mathematics

- **GPA:** 3.91/4.0.
- **Honors:** C.P. Davis Scholar in Columbia Engineering, Dean's List (all semesters).
- **Coursework:** ODEs, Discrete Math, Linear Algebra, Advanced Programming, Data Structures, Probability.
- **Activities:** Columbia Space Initiative, Formula SAE, Columbia Quant Group.

The University of Chicago Laboratory High School **Chicago, IL**

- **Relevant Coursework:** Linear Algebra and Multivariable Calculus, Computer Architecture, AP Physics C.
- **Notable Activities:** FIRST Robotics Competition #8096 (Co-Captain, Software Lead, Pit Crew Lead).
- **The University of Chicago Master's Program in Computer Science (Dual Enrollment)** — Unix Systems.

PROGRAMMING LANGUAGES & SKILLS

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- **Programming Languages:** Python (libraries: pandas, NumPy, openpyxl, pytest, Playwright, altair; APIs: Bloomberg, Microsoft Graph, Claude, Gmail), Java (libraries: WPI Robotics Library, PathPlannerLib), Bash, C.
 - **Related Skills:** Command Line Interface, Git, LaTeX, Palantir AIP.